

1 Models

BarotropicQG, ThermalQG, and ThermalQG_coupled classes are instances of the parent class QG_model. Their methods are therefore similar. beta (β -plan approximation) is the only common attributes.

1.1 Common functions and attributes

- `psi_from_PV(PV, assign = False)`
- `PV_from_psi(psi, assign = False)`
- `vort_from_PV(PV=None)`
- `PV_from_vort(vort = None, assign = False)`
- `psi_from_U(U, constant = 0.0)`
- `vort_from_U(U)`

1.2 Barotropic QG (class |BarotropicQG|)

1.2.1 Equations

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi - \frac{\psi}{R_d^2} = q - \beta y \quad (1.1)$$

$$\frac{\partial q}{\partial t} = -J(\psi, q) \quad (1.2)$$

ψ : streamfunction, q : potential vorticity. Fields variables :

- streamfunction : $\psi = \psi$
- potential vorticity : $PV = q$:
- relative vorticity : $\text{vorticity} = \zeta = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi = q - \beta y + \frac{\psi}{R_d^2}$

1.2.2 Diagnostics

- Total potential vorticity : $PV_total = Q = \iint_S q(x, y, t) dx dy$
- Kinetic energy : $\text{kinetic_energy} = E_c = \frac{1}{2} \iint_S \left(\left(\frac{\partial \psi}{\partial x} \right)^2 + \left(\frac{\partial \psi}{\partial y} \right)^2 \right) dx dy$
- Available potential energy : $\text{potential_energy} = E_p = \frac{1}{2} \iint_S \frac{\psi^2}{R_d^2} dx dy$
- Total energy : $\text{total_energy} = E = E_c + E_p$
- Enstrophy : $\text{enstrophy} = Z = \frac{1}{2} \iint_S q^2 dx dy$

1.2.3 Parameters

- Rossby Radius : $\text{inv_Rd2} = \frac{1}{R_d^2}$
- Mean zonal transport (for zonal channel configuration) : $T_0 = T_0$

1.3 Thermal Quasi-Geostrophic (class ThermalQG)

Formulation based on Warneford and Dellar, 2013

1.3.1 Equations

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi - \frac{\psi}{R_d^2} = q - \frac{\theta}{R_d^2} - \beta y \quad (1.3)$$

$$\frac{\partial q}{\partial t} = -J(\psi, q) + \frac{1}{R_d^2} J(\psi, \theta) \quad (1.4)$$

$$\frac{\partial \theta}{\partial t} = -J(\psi, \theta) \quad (1.5)$$

ψ : streamfunction, q : potential vorticity, θ : buoyancy. Fields variables :

- streamfunction : $\psi = \psi$
- buoyancy : $\theta = \theta$
- potential vorticity : $PV = q$
- relative vorticity : $\zeta = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi = q - \beta y + \frac{\psi - \theta}{R_d^2}$

1.3.2 Diagnostics

- Total potential vorticity : $PV_total = Q = \iint_S q(x, y, t) dx dy$
- Kinetic energy : $kinetic_energy = E_c = \frac{1}{2} \iint_S \left(\left(\frac{\partial \psi}{\partial x} \right)^2 + \left(\frac{\partial \psi}{\partial y} \right)^2 \right) dx dy$
- Available potential energy : $potential_energy = E_p = \frac{1}{2} \iint_S \frac{\psi^2}{R_d^2} dx dy$
- Total energy : $total_energy = E = E_c + E_p$
- Total buoyancy : $theta_total = \Theta = \iint_S \theta(x, y, t) dx dy$
- Enstrophy : $enstrophy = Z = \frac{1}{2} \iint_S q^2 dx dy$

1.3.3 Parameters

- Rossby Radius : $inv_Rd2 = \frac{1}{R_d^2}$
- Mean zonal transport (for zonal channel configuration) : $T_0 = T_0$

1.4 Coupled Thermal Quasi-Geostrophic (class ThermalQG_coupled)

1.4.1 Equations

1.4.2 Diagnostics

1.4.3 Parameters

2 General parameters (class Params)

3 Numeric

4 Time schemes

We solve ODE like this

$$\frac{du(t)}{dt} = \text{RHS}(u, t) \quad (4.1)$$

$$u(0) = u^0 \quad (4.2)$$

Time Step is noted Δt .

Euler

$$u^{n+1} = u^n + \Delta t \times \text{RHS}(u^n, t^n) \quad (4.3)$$

RK2

$$u^{n+1} = u^n + \frac{\Delta t}{2}(k_1 + k_2) \quad (4.4)$$

$$k_1 = \text{RHS}(u^n, t^n) \quad (4.5)$$

$$k_2 = \text{RHS}(u^n + \Delta t k_1, t^n + \Delta t) \quad (4.6)$$

RK4

$$u^{n+1} = u^n + \frac{\Delta t}{3} \left(\frac{k_1}{2} + k_2 + k_3 + \frac{k_4}{2} \right) \quad (4.7)$$

$$k_1 = \text{RHS}(u^n, t^n) \quad (4.8)$$

$$k_2 = \text{RHS}\left(u^n + \frac{\Delta t}{2} k_1, t^n + \frac{\Delta t}{2}\right) \quad (4.9)$$

$$k_3 = \text{RHS}\left(u^n + \frac{\Delta t}{2} k_2, t^n + \frac{\Delta t}{2}\right) \quad (4.10)$$

$$k_4 = \text{RHS}(u^n + \Delta t k_3, t^n + \Delta t) \quad (4.11)$$

Leap-Frog

$$u^{n+1} = u^{n-1} + 2\Delta t \text{RHS}(u^n, t^n) \quad (4.12)$$

First step is computed with Euler scheme. To remove computationnal mode, we use Asselin filter :

$$u^{n+1} = u^{n-1} + 2\Delta t \text{RHS}(\bar{u}^n, t^n) \quad (4.13)$$

$$\bar{u}^n = u^n + \alpha(u^{n+1} + 2u^n + u^{n-1}) \quad (4.14)$$

$$\alpha \approx 0.05 \quad (4.15)$$

References

Warneford, E. S., & Dellar, P. J. (2013). The quasi-geostrophic theory of the thermal shallow water equations. *Journal of Fluid Mechanics*, 723, 374–403. <https://api.semanticscholar.org/CorpusID:124223095>